

Haokai Ding

Shenzhen, China

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Summary

Undergraduate researcher in robotics, focusing on underactuated grasping, idle-stroke and Peaucellier-based linkages, and contact-rich aerial manipulation. Building hardware prototypes and analytical models that bridge mechanism design with field-deployable manipulation. Incoming MSc in Robotics at MBZUAI.

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

M.Sc. IN ROBOTICS (INCOMING)

Abu Dhabi, UAE

Fall 2026

Tsinghua University | Shenzhen X-Institute

TSIEN EXCELLENCE IN ENGINEERING PROGRAM (TEEP) – X SCHOLAR

Shenzhen, China

Sep. 2023 – Present

- Joint training program in underactuated robotics and mechanism design.

Shenzhen Technology University – Future Technology School

B.ENG. IN ELECTRONIC SCIENCE AND TECHNOLOGY

Shenzhen, China

Sep. 2022 – Jul. 2026

- GPA: 84.6/100. Research focus: LIBS spectroscopy and robotics systems.

Research Experience

State Key Lab of Mechanical Systems & Vibration, Shanghai Jiao Tong University

VISITING STUDENT – UAV PERCHING AND AERIAL MANIPULATION

Shanghai, China

Feb. 2026 – Present

- Advisor: Prof. Wei Dong.
- Working on UAV perching and contact-rich aerial manipulation.

Shenzhen X-Institute, Tsinghua University

UNDERGRADUATE RESEARCHER – UNDERACTUATED ROBOTIC GRIPPERS

Shenzhen, China

2023 – Present

- Designed and tested underactuated grippers based on semi-Peaucellier linkages.
- Developed idle-stroke and delay-triggering mechanisms; validated prototypes via simulation and physical experiments.

Shenzhen Technology University

UNDERGRADUATE RESEARCHER – LIBS SPECTROSCOPY FOR HEAVY-METAL ANALYSIS

Shenzhen, China

2022 – 2023

- Built and optimized a LIBS system for rapid heavy-metal adsorption analysis.
- Improved the calibration pipeline and acquisition accuracy.

Publications

† Co-first author * Corresponding author Listed in reverse chronological order by IEEE Xplore date added.

SELECTED CONFERENCE PAPERS (ROBOTICS)

Y. Chen, H. Luan, **H. Ding**, W. Zhang*

IEEE RAAI 2025, Singapore

[P1] AN IDLE-STROKE UNDERACTUATED GRIPPER WITH INDEPENDENT DOUBLE-PARALLELOGRAM LINKAGES FOR PINCHING AND ADAPTIVE GRASPING

Mar. 2026

H. Ding†, X. Zhong, W. Huang, T. Yang*, W. Zhang*

IEEE ROBIO 2025, Bangkok, Thailand

[P2] BioCREST – A FLEXIBLE ADAPTIVE ROBOTIC HAND BASED ON IDLE-STROKE MECHANISM

Feb. 2026

H. Ding, H. Luan, T. Yang*, W. Zhang*

IEEE ICCMA 2025, Macau, China

[P3] GLINT: AN IDLE-STROKE GRASP-AND-LIFT HAND FOR IN-HAND MANIPULATION

Feb. 2026

H. Ding†, J. Fan, Z. Zhu, Y. Zhao, K. Chen*, W. Zhang*

IEEE ICARM 2025, Hangzhou, China

[P4] PEAUCELLIER GRIPPER: A NOVEL UNDERACTUATED GRIPPER FOR LINEAR PINCHING AND SELF-ADAPTIVE GRASP

Dec. 2025

H. Ding†, Y. Chen†, D. Jia, W. Zhang*

IEEE ICIA 2025, Wuhan, China

[P5] AN UNDERACTUATED GRIPPER WITH GRASSHOPPER-INSPIRED LINKAGES AND DELAY TRIGGERING FOR PINCHING AND ADAPTIVE GRASPING

Dec. 2025

H. Ding[†], W. Zhang^{*}

*IEEE/RSJ IROS 2025, Hangzhou,
China*

[P6] A NOVEL GRIPPER WITH SEMI-PEAUCELLIER LINKAGE AND IDLE-STROKE MECHANISM FOR LINEAR PINCHING AND SELF-ADAPTIVE GRASPING

Nov. 2025

H. Ding[†], Z. Chen, T. Yang^{*}, W. Zhang^{*}

IEEE CASE 2025, Los Angeles, USA

[P7] SEMI-PEAUCELLIER LINKAGE AND DIFFERENTIAL MECHANISM FOR LINEAR PINCHING AND SELF-ADAPTIVE GRASPING

Sep. 2025

OTHER PUBLICATIONS

Y. Liu[†], H. Ding[†], H. Chen, H. Gao, J. Yu, F. Mo, N. Wang^{*}

Polymers (MDPI) 17(15):2136

[P8] RECENT PROGRESS ON THE RESEARCH OF 3D PRINTING IN AQUEOUS ZINC-ION BATTERIES

Aug. 2025

H. Xie, L. You, X. Fang, L. Li, H. Ding, Z. Zhou, G. Zhang, D. Zhang

E3S Web of Conferences, 615 (2025)

02010

[P9] RAPID ANALYSIS OF HEAVY-METAL ELEMENT ADSORPTION BY SCG BASED ON LIBS TECHNOLOGY

Feb. 2025

Honors & Awards

INTERNATIONAL AWARDS

2025 **Undergraduate Champion**, ASME Student Mechanism & Robot Design Competition

USA

NATIONAL & DOMESTIC AWARDS

2025 **National Scholarship**, Ministry of Education of China

China

2024 **President's Award**, Shenzhen Technology University

Shenzhen, China

2024 **Research and Innovation Award (First Prize)**, Shenzhen Technology University

Shenzhen, China

2023 **Gold Prize**, 9th China "Internet+" Innovation Competition (Guangdong Division)

Guangdong, China

2023 **Second Prize**, 17th "Challenge Cup" Extracurricular Science Competition (Guangdong Division)

Guangdong, China

2023 **Research and Innovation Award (First Prize)**, Shenzhen Technology University

Shenzhen, China

Professional Service

2025 **Conference Reviewer**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

2025 **Session Chair**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Hangzhou, China

2025 **Conference Reviewer**, IEEE International Conference on Automation Science and Engineering (CASE)